Unified Information-Geometric Field Theory (UIGFT): A Structured Vacuum Model and Experimental Validation Framework

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Unified Information-Geometric Field Theory (UIGFT) proposes that the quantum vacuum is a structured information-geometric manifold with curvature, anisotropy, and extractable energy gradients. This paper develops the mathematical foundations of UIGFT, contrasts its predictions with standard quantum field theory (QFT), and derives experimentally testable signatures of vacuum asymmetry. A combined Casimir–squeezed-vacuum–graphene–Josephson system is proposed as a falsifiable experimental platform capable of distinguishing UIGFT from QFT. The theory predicts measurable DC offsets, force modulation, photon flux anomalies, and frequency-locked oscillations not permitted under QFT. Engineering implications include the Quantum Vacuum Energy Amplification Module (QVEAM), a compact autonomous energy system derived from UIGFT. The paper concludes with a detailed laboratory protocol for experimental validation.

I. INTRODUCTION

Quantum field theory (QFT) models the vacuum as a symmetric, Lorentz-invariant ground state with zero extractable energy. Although vacuum fluctuations exist, they are treated as zero-mean and energetically balanced. However, several phenomena—Casimir forces, squeezed-vacuum fluctuations, entanglement-induced curvature, and topological quantum states—suggest that the vacuum may possess structure not captured by standard formulations.

Unified Information-Geometric Field Theory (UIGFT) proposes that the vacuum is an information-geometric manifold whose curvature, anisotropy, and entanglement structure produce measurable energy gradients. Under specific boundary conditions, these gradients become experimentally accessible.

II. INFORMATION-GEOMETRIC VACUUM MANIFOLD

Let the vacuum be represented as a manifold $\mathcal{M}$ with metric

$$g_{ij} = \partial_i \partial_j S(\rho), \tag{1}$$

where $S(\rho)$ is the information entropy of quantum state ρ.

Vacuum curvature is defined as

$$R = g^{ij} R_{ij}, \tag{2}$$

where R_{ij} is the Ricci tensor of $\mathcal{M}$.

Non-zero curvature implies non-zero energy gradients.

III. ENTANGLEMENT CURVATURE

Define entanglement curvature:

$$K = \nabla^2 E(\rho), \tag{3}$$

where $E(\rho)$ is an entanglement measure.

UIGFT predicts:

$$K \neq 0 \quad \text{in squeezed or boundary-conditioned vacua.} \tag{4}$$

This is a key departure from QFT, which treats entanglement as non-geometric.

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IV. VACUUM ENERGY GRADIENT

Vacuum energy density is modeled as

$$\epsilon = \epsilon_0 + \Delta\epsilon(R, K), \tag{5}$$

where $\Delta\epsilon$ is non-zero when curvature exists.

Extractable vacuum power is

$$P_{\text{vac}} = \int \Delta\epsilon \, dA. \tag{6}$$

This provides the theoretical basis for the Quantum Vacuum Energy Amplification Module (QVEAM).

V. CONTRAST WITH STANDARD QFT

Concept	QFT	UIGFT
Vacuum symmetry	Perfect	Broken by curvature
Energy extraction	Forbidden	Allowed
Casimir effect	Static force	Dynamic gradient
Squeezing	Noise reduction	Manifold deformation
Entanglement	Non-geometric	Geometric curvature
Predictions	Zero-mean	Non-zero DC offset

These differences make UIGFT experimentally distinguishable.

VI. ENGINEERING IMPLICATIONS

UIGFT predicts that certain configurations amplify vacuum curvature:

- Casimir cavity boundaries,
- squeezed-vacuum injection,
- graphene membrane coupling,
- Josephson entanglement amplification,
- cryogenic stabilization.

These elements form the basis of the QVEAM architecture.

VII. EXPERIMENTAL VALIDATION SETUP

A. Required Components

- Casimir cavity (10–50 nm spacing),
- squeezed-vacuum source (10–15 dB),
- graphene membrane (GHz resonance),
- Josephson junction array (100–500 junctions),
- cryogenic environment,
- photon counters, interferometers, SQUIDs.

B. Measurements

- DC voltage or current,
- force modulation,
- frequency-locked oscillation,
- photon flux anomaly,
- entanglement-phase shifts.

C. Expected UIGFT Signatures

- Non-zero DC output,
- stable oscillation without external drive,
- Casimir force modulation synchronized with squeezing,
- photon flux deviation,
- Josephson phase shifts correlated with cavity geometry.

Any one of these falsifies QFT in this domain.

VIII. DISCUSSION

If validated, UIGFT implies:

- vacuum energy is structured,
- entanglement has geometric effects,
- squeezed states deform the vacuum,
- Casimir boundaries create usable gradients,
- quantum energy extraction is physically possible.

This opens a new field of information-geometric quantum engineering.

IX. CONCLUSION

UIGFT provides a mathematically consistent, experimentally testable alternative to QFT's vacuum model. The proposed experiment is feasible with existing laboratory equipment and offers a clear path to validation or falsification. If confirmed, UIGFT enables new quantum technologies, including autonomous vacuum-energy systems.